# Smooth lattice 3-polytopes with seven facets have the integer decomposition property

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**Abstract**

We prove Oda's conjecture unconditionally for smooth lattice polytopes in dimension three with seven facets: every such polytope has the integer decomposition property. With the known case of at most six facets, the conjecture holds for every smooth lattice 3-polytope with at most seven facets.

**Mathematics Subject Classifications:** Primary 52B20; Secondary 14M25, 68V15

## 1 Introduction

A *lattice polytope* $P \subset \mathbb{R}^n$ is the convex hull of finitely many points of $\mathbb{Z}^n$. It has the *integer decomposition property* (IDP) if for every integer $k \geqslant 1$ every point of $kP \cap \mathbb{Z}^n$ is a sum of $k$ points of $P \cap \mathbb{Z}^n$. It is *smooth* if it is simple and at every vertex the primitive edge directions form a basis of $\mathbb{Z}^n$; equivalently, its normal fan is smooth. Smooth lattice polytopes are the polytopes of ample line bundles on smooth projective toric varieties, and the IDP of such a polytope is the projective normality of the corresponding embedding.

Oda asked whether every smooth lattice polytope has the IDP [16, Problems 2 and 3]; the affirmative answer is known as Oda's conjecture. Every lattice polygon has the IDP (Lemma 3), so the conjecture holds in dimension two. In dimension three it is open. It holds for smooth 3-polytopes that are centrally symmetric [1], that have at most six facets [19] (more generally, for smooth complete fans in $\mathbb{R}^n$ with at most $n+3$ rays), and that are pseudo-prisms [5]. It also holds for the polytope of an ample line bundle $L$ on a smooth projective toric threefold $X$ when the adjoint bundle $L + K_X$ has no nonzero section or is not big [17], and then for the polytopes of all ample line bundles on $X$ [18]; this covers the toric Fano threefolds. In every dimension it holds for smooth combinatorial cubes

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[5] and for smooth polytopes combinatorially equivalent to products of simplices [6]. We prove the next case in dimension three.

**Theorem 1.** *Every smooth lattice polytope in $\mathbb{R}^3$ with seven facets has the integer decomposition property.*

**Corollary 2.** *Every smooth lattice polytope in $\mathbb{R}^3$ with at most seven facets has the integer decomposition property.*

The proof has four parts, shown in Figure 1. By Lemma 3, a lattice 3-polytope $P$ has the IDP as soon as $2P \cap \mathbb{Z}^3 = (P \cap \mathbb{Z}^3) + (P \cap \mathbb{Z}^3)$. Propositions 6, 7 and 8 give three conditions on the normal fan of $P$, each of which implies this equality: a slicing condition and two projection conditions. Proposition 9 lists the smooth complete fans with seven rays that are normal fans of polytopes, up to $\mathrm{GL}_3(\mathbb{Z})$: they form 625 families whose rays are affine functions of integer parameters. Proposition 10 gives for each family a vector that satisfies one of the three conditions for all values of the parameters. For 25 families this is proved in the text; for the remaining 600 families it is a finite computation with polynomials in the parameters, which is formalized in Lean 4 and checked by its kernel for all 625 families. The conditions depend only on the normal fan, so the theorem holds for every polytope with a given normal fan at once.

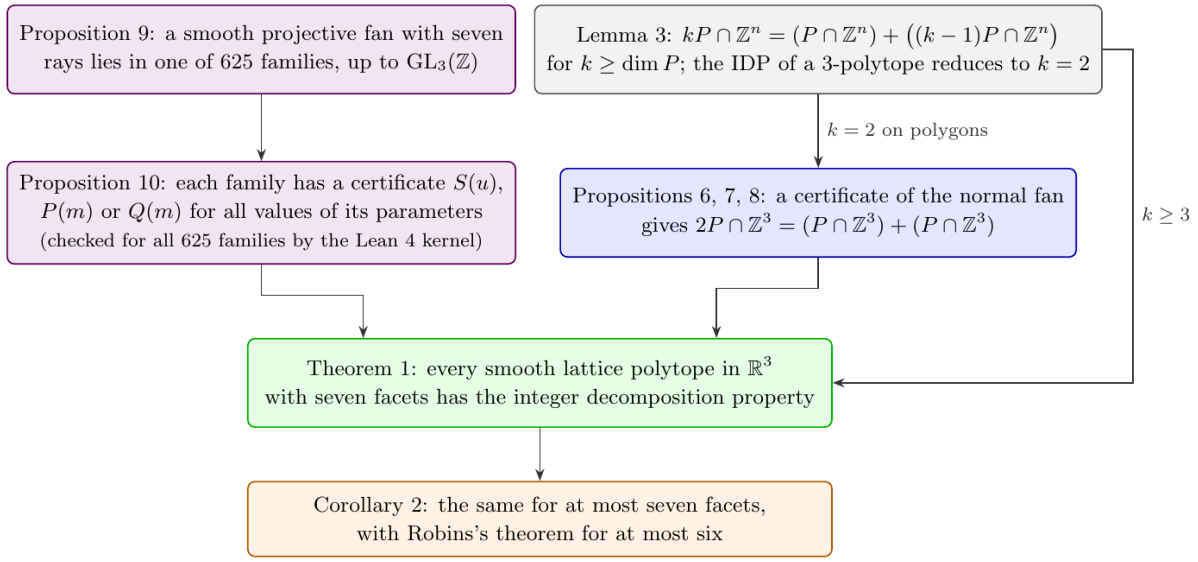



Figure 1: The structure of the proof. Lemma 3 reduces the IDP to the second dilate, Propositions 6 to 8 turn a certificate of the normal fan into the equality for the second dilate, and Propositions 9 and 10 supply a certificate for every smooth projective fan with seven rays.

## 2 Reduction to the second dilate

**Lemma 3.** *Let $P \subset \mathbb{R}^n$ be a lattice polytope of dimension $d$ and let $k \geqslant \max(d, 1)$ be an integer. Then*

$$kP \cap \mathbb{Z}^n = (P \cap \mathbb{Z}^n) + \big((k-1)P \cap \mathbb{Z}^n\big).$$

*In particular every lattice polygon has the IDP, and a lattice 3-polytope $P$ has the IDP if and only if $2P \cap \mathbb{Z}^3 = (P \cap \mathbb{Z}^3) + (P \cap \mathbb{Z}^3)$.*

This is the argument behind [3, Theorem 1.3.3] and [9]; we include it because it is used in every proof below.

*Proof.* The inclusion $\supseteq$ is clear. If $d = 0$ then $P = \{v\}$ and $x = kv = v + (k-1)v$. Let $d \geqslant 1$ and fix a triangulation $T$ of $P$ whose vertices are exactly the points of $P \cap \mathbb{Z}^n$ (for instance a placing triangulation [7]). A simplex $S \in T$ contains no lattice point other than its vertices, since such a point would be a vertex of $T$ lying in $S$.

Let $x \in kP \cap \mathbb{Z}^n$. Then $x \in kS$ for a $d$-simplex $S \in T$ with vertices $v_0, \ldots, v_d$, and $x = \sum_i \lambda_i v_i$ with $\lambda_i \geqslant 0$ and $\sum_i \lambda_i = k$. Write $\lambda_i = \lfloor \lambda_i \rfloor + \mu_i$ with $0 \leqslant \mu_i < 1$. The point $y = \sum_i \mu_i v_i = x - \sum_i \lfloor \lambda_i \rfloor v_i$ lies in $\mathbb{Z}^n$, and $\sum_i \mu_i = k - \sum_i \lfloor \lambda_i \rfloor$ is an integer smaller than $d+1$. If $\sum_i \mu_i = d$, then every $\mu_i$ is positive and $\sum_i v_i - y = \sum_i (1-\mu_i) v_i$ is a lattice point of $S$ with all barycentric coordinates in $(0, 1)$, which is not a vertex of $S$; this is impossible. Hence $\sum_i \mu_i \leqslant d - 1 < k$, so $\lfloor \lambda_j \rfloor \geqslant 1$ for some $j$. Then $x = v_j + (x - v_j)$, where $x - v_j = \sum_i (\lambda_i - \delta_{ij}) v_i$ has nonnegative coefficients with sum $k-1$, so $x - v_j \in (k-1)S \cap \mathbb{Z}^n \subseteq (k-1)P \cap \mathbb{Z}^n$.

For a polygon the case $k \geqslant 2$ gives the IDP by induction on $k$. For a 3-polytope, the case $k \geqslant 3$ shows $kP \cap \mathbb{Z}^3 = (P \cap \mathbb{Z}^3) + ((k-1)P \cap \mathbb{Z}^3)$, and induction on $k$ reduces the IDP to $k = 2$. □

The lemma applies to any full-rank lattice $\Lambda \subset \mathbb{R}^n$ in place of $\mathbb{Z}^n$, by a linear change of coordinates, and to a polytope whose vertices lie in a coset $w + \Lambda$, by translation by $w$. We use it in this form for polygons and segments in planes of $\mathbb{R}^3$.

## 3 Three criteria

**Notation.** We use the standard scalar product $\langle x, y \rangle$ on $\mathbb{R}^3$, the determinant $\det(a, b, c)$ of three vectors, and the cross product, for which $\langle a \times b, w \rangle = \det(a, b, w)$. A lattice 3-polytope is written irredundantly as

$$P = \{x \in \mathbb{R}^3 : \langle x, \rho_i \rangle \geqslant c_i \ (0 \leqslant i < r)\}$$

with primitive $\rho_i \in \mathbb{Z}^3$; then $c_i \in \mathbb{Z}$. Let $\Phi_i = \{x \in P : \langle x, \rho_i \rangle = c_i\}$ be the facets. The normal fan $\Sigma$ of $P$ has rays $\rho_0, \ldots, \rho_{r-1}$; its two-dimensional cones $\operatorname{cone}(\rho_i, \rho_j)$ correspond to the edges $\Phi_i \cap \Phi_j$, and its maximal cones to the vertices. If $P$ is smooth, every maximal cone $\operatorname{cone}(\rho_i, \rho_j, \rho_k)$ is generated by a basis of $\mathbb{Z}^3$, and the edges at the corresponding vertex $v$ point in the directions of the dual basis $e_i, e_j, e_k$, where $\langle e_i, \rho_i \rangle = 1$ and $\langle e_i, \rho_j \rangle = \langle e_i, \rho_k \rangle = 0$.

**Lemma 4.** *(a) Let $\alpha, \beta \in \mathbb{Z}^2$ and $a, b \in \mathbb{Z}$, and suppose the lines $\langle y, \alpha\rangle = a$ and $\langle y, \beta\rangle = b$ in $\mathbb{R}^2$ meet in exactly one point $q$. If $|\det(\alpha, \beta)| = 1$ then $q \in \mathbb{Z}^2$.*

*(b) Let $A, B \subset \mathbb{R}^2$ be convex polygons and let $q$ be a vertex of $A \cap B$ that is a vertex of neither $A$ nor $B$. Then $q$ lies in the relative interior of an edge of $A$ and in the relative interior of an edge of $B$, and the lines spanned by these two edges are distinct.*

*Proof.* (a) By Cramer's rule $q$ is the image of $(a, b)$ under the inverse of an integer matrix of determinant $\pm 1$. (b) If $q$ were an interior point of $A$, then $A \cap B$ would coincide with $B$ near $q$ and $q$ would be a vertex of $B$. So $q \in \partial A$, and as $q$ is not a vertex of $A$ it lies in the relative interior of an edge of $A$; similarly for $B$. If the two edges spanned the same line, then near $q$ both $A$ and $B$ would be half-planes bounded by this line, and $q$ would not be a vertex of $A \cap B$. □

We also use the following result of Haase, Nill, Paffenholz and Santos.

**Lemma 5** ([11, Proposition 2.1])**.** *Let $A, B \subset \mathbb{R}^2$ be two-dimensional lattice polygons such that $A \cap B$ is nonempty and contains no point of $\mathbb{Z}^2$. Then $A \cap B$ is a quadrilateral, two opposite edges of which lie in edges of $A$ and the other two in edges of $B$.*

## Slicing

**Proposition 6.** *Let $P$ be a smooth lattice 3-polytope with normal fan $\Sigma$, and let $u \in \mathbb{Z}^3 \setminus \{0\}$ satisfy:*

*(S1) for every maximal cone $\mathrm{cone}(\rho_i, \rho_j, \rho_k)$ of $\Sigma$, $u = a\rho_i + b\rho_j + c\rho_k$ with $a, b, c \in \{-1, 0, 1\}$;*

*(S2) the graph $G_u$ on $\{0, \dots, r-1\}$ in which $i$ and $j$ are adjacent when $|\det(u, \rho_i, \rho_j)| \geqslant 2$ contains no cycle of length four.*

*Then $2P \cap \mathbb{Z}^3 = (P \cap \mathbb{Z}^3) + (P \cap \mathbb{Z}^3)$.*

*Proof.* By (S1) $u$ is primitive, so after a change of coordinates by $\mathrm{GL}_3(\mathbb{Z})$, which changes all determinants by the same sign, we may assume $u = (0, 0, 1)$. For $x \in \mathbb{R}^3$ write $x = (x', x_3)$ with $x' \in \mathbb{R}^2$, and for each ray write $\rho_i = (\bar{\rho}_i, \rho_i^3)$. Then $\det(u, \rho_i, \rho_j) = \det(\bar{\rho}_i, \bar{\rho}_j)$. For $t \in \mathbb{R}$ let $L_t = \{x_3 = t\}$ and $P_t = P \cap L_t$, and let $[t_-, t_+]$ be the image of $P$ under $x \mapsto x_3$; its ends are integers. We identify $L_t$ with $\mathbb{R}^2$ by $x \mapsto x'$; for integer $t$ this identifies $L_t \cap \mathbb{Z}^3$ with $\mathbb{Z}^2$. On $L_t$ the facet $\Phi_k$ lies in the line $\langle x', \bar{\rho}_k\rangle = c_k - t\rho_k^3$, whose normal $\bar{\rho}_k$ and right-hand side are integral when $t \in \mathbb{Z}$.

*Step 1: for $t \in \mathbb{Z}$, $P_t$ is a lattice polytope.* A vertex $q$ of $P_t$ is either a vertex of $P$ or the intersection of $L_t$ with an edge of $P$ not contained in $L_t$. In the second case the edge starts at a vertex $v$ of $P$ in a direction $e$ of the dual basis of the maximal cone of $v$, and $\langle e, u\rangle = e_3$ is the coefficient of $u$ at the corresponding ray, which is $\pm 1$ by (S1). Hence $q = v \pm (t - v_3)e \in \mathbb{Z}^3$.

*Step 2: for $t \in \mathbb{Z}$ with $t < t_+$, $P_{t+1/2} = \frac{1}{2}(P_t + P_{t+1})$.* All vertices of $P$ lie at integer heights, so the polytope $P \cap \{t \leqslant x_3 \leqslant t+1\}$ has all its vertices in $L_t \cup L_{t+1}$ and equals $\operatorname{conv}(P_t \cup P_{t+1})$. Its section at height $t + \frac{1}{2}$ is $\frac{1}{2}(P_t + P_{t+1})$.

*Step 3: edges of the slices.* If $t_- < t < t_+$, then $L_t$ meets the interior of $P$ and $P_t$ is a polygon. If $E$ is an edge of $P_t$, a point of $\operatorname{relint} E$ lies on $\partial P$, hence on a facet $\Phi_k$; then $\Phi_k \cap L_t$ is a face of $P_t$ that contains this point and is not $P_t$, so $\Phi_k \cap L_t = E$ and $\bar{\rho}_k \neq 0$. If $t \in \{t_-, t_+\}$, then $P_t$ is a face of $P$. If it is a facet $\Phi_q$, every edge of $\Phi_q$ is $\Phi_q \cap \Phi_k = \Phi_k \cap L_t$ for a neighbouring facet $\Phi_k$, and $\bar{\rho}_k \neq 0$. If it is an edge $\Phi_i \cap \Phi_j$, then $x \mapsto x_3$ attains its minimum or its maximum on $P$ exactly on this edge, so $\pm u$ lies in the relative interior of $\operatorname{cone}(\rho_i, \rho_j)$; by (S1) applied to a maximal cone containing $\rho_i$ and $\rho_j$ we get $u = \pm(\rho_i + \rho_j)$, hence $\bar{\rho}_j = -\bar{\rho}_i \neq 0$, and the segment $P_t$ lies on the line $\langle x', \bar{\rho}_i \rangle = c_i - t\rho_i^3$, which is also the line $\langle x', \bar{\rho}_j \rangle = c_j - t\rho_j^3$.

In all cases we call $k$ a *label* of the edge $E = \Phi_k \cap L_t$ of a polygon $P_t$; different edges of $P_t$ have different labels.

*Step 4: even heights.* Let $z \in 2P \cap \mathbb{Z}^3$ with $z_3 = 2t$. Then $z/2 \in P_t$, a lattice polytope of dimension at most 2 in $L_t$ by Step 1, and Lemma 3 with $k = 2$ writes $z$ as a sum of two lattice points of $P_t$.

*Step 5: odd heights.* Let $z \in 2P \cap \mathbb{Z}^3$ with $z_3 = 2t + 1$. By Step 2, $z \in P_t + P_{t+1}$, so $A = P_t$ and $B = z - P_{t+1}$ are lattice polytopes in $L_t$ with $A \cap B \neq \emptyset$. A lattice point $x \in A \cap B$ gives $z = x + (z - x)$ with $x, z - x \in P \cap \mathbb{Z}^3$, so it suffices to show that $A \cap B$ contains a lattice point. The edges of $B$ are the sets $z - E$ for edges $E$ of $P_{t+1}$; such an edge with label $k$ lies on a line $\langle x', \bar{\rho}_k \rangle = b$ with $b \in \mathbb{Z}$, and we give $z - E$ the label $k$. Suppose that $A \cap B$ contains no lattice point.

If $A$ or $B$ is a point, it is a lattice point of $A \cap B$. If $A$ and $B$ are polygons, Lemma 5 shows that $A \cap B$ is a quadrilateral whose sides lie, in cyclic order, in edges of $A$, $B$, $A$, $B$ with labels $i, j, k, l$. Two opposite sides of a quadrilateral are not collinear, so $i \neq k$ and $j \neq l$. No vertex of the quadrilateral is a vertex of $A$ or $B$, so by Lemma 4(b) each vertex is the only common point of the lines of two consecutive sides, and by Lemma 4(a) these lines have normals with $|\det| \geqslant 2$. Thus $i \sim j \sim k \sim l \sim i$ in $G_u$. Adjacent vertices of $G_u$ are distinct, so this is a cycle of length four, contrary to (S2).

If $A$ is a segment, then by Step 3 $t = t_\pm$, $A = \Phi_i \cap \Phi_j$ and $\bar{\rho}_j = -\bar{\rho}_i$. Suppose first that $B$ is a segment too. Then $t+1$ is also an end of $[t_-, t_+]$, $P_{t+1} = \Phi_k \cap \Phi_l$ and $\bar{\rho}_l = -\bar{\rho}_k$. If $A$ and $B$ were collinear, an end of the segment $A \cap B$ would be an end of $A$ or of $B$, a lattice point. So $A$ and $B$ meet in a single point of their relative interiors, and Lemma 4(a) gives $|\det(\bar{\rho}_i, \bar{\rho}_k)| \geqslant 2$. As $\bar{\rho}_j = -\bar{\rho}_i$ and $\bar{\rho}_l = -\bar{\rho}_k$, the same holds for the pairs $(i, l)$, $(j, k)$, $(j, l)$, and $i, k, j, l$ is a cycle of length four in $G_u$. Suppose now that $B$ is a polygon. An end $q$ of $A \cap B$ is not an end of $A$ and not a vertex of $B$, so it lies in $\operatorname{relint} A$ and in the relative interior of an edge of $B$ with some label $k$. If $A$ were contained in the line of this edge, the ends of $A \cap B$ would be lattice points. So $A$ crosses this line at $q$ and enters the interior of $B$ there; hence $A \cap B$ is a segment $[q, q']$ with $q \neq q'$, and $q'$ lies on an edge of $B$ with a label $l \neq k$. Lemma 4(a) gives $i \sim k$ and $i \sim l$, and $\bar{\rho}_j = -\bar{\rho}_i$ gives $j \sim k$ and $j \sim l$: the cycle $i, k, j, l$. The case where $B$ is a segment and $A$ a polygon is the same with $A$ and $B$ exchanged. Each case contradicts (S2). □

Figure 2 shows Steps 2 and 5 of the proof on a small example.

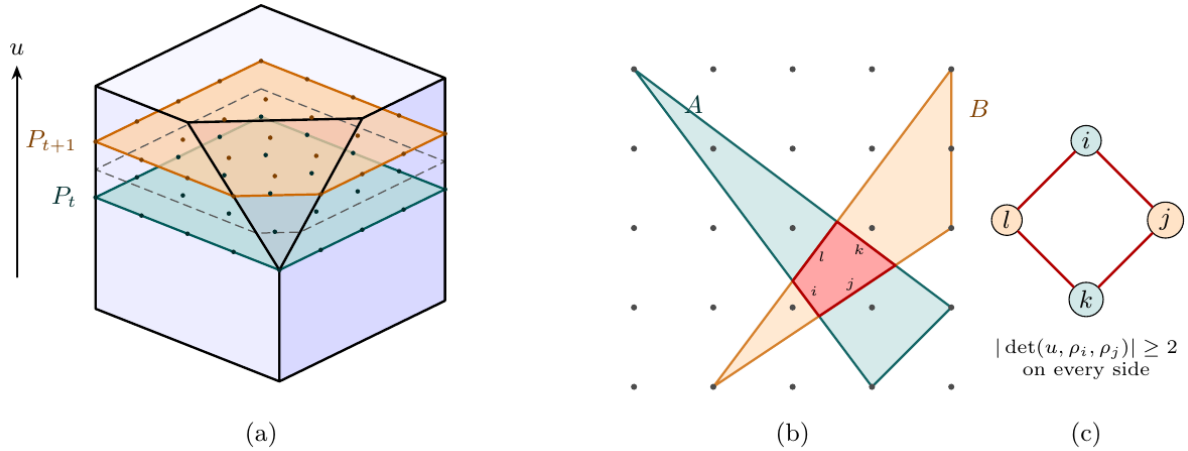



Figure 2: The slicing criterion. (a) The slices $P_t$ and $P_{t+1}$ of a smooth polytope at consecutive integer heights; the section at height $t + \frac{1}{2}$ is their average. (b) Two lattice polygons $A$ and $B$ whose intersection contains no lattice point: it is a quadrilateral whose sides lie alternately on edges of $A$ and of $B$. (c) The labels of the four sides form a cycle of length four in $G_u$, which condition (S2) excludes.

## Projection

For Propositions 7 and 8 let $P$ be a lattice 3-polytope, $m \in \mathbb{Z}^3$ primitive, and $h_i = \langle \rho_i, m \rangle$ the *height* of $\rho_i$. Choose $b_1, b_2 \in \mathbb{Z}^3$ such that $b_1, b_2, m$ is a basis of $\mathbb{Z}^3$, write points as $x = y_1 b_1 + y_2 b_2 + sm$ with $y = (y_1, y_2)$, and put $\rho_i' = (\langle b_1, \rho_i \rangle, \langle b_2, \rho_i \rangle)$. Then $\langle x, \rho_i \rangle = \langle y, \rho_i' \rangle + s h_i$, and $x \in \mathbb{Z}^3$ if and only if $y \in \mathbb{Z}^2$ and $s \in \mathbb{Z}$. Facets with $h_i > 0$ are *lower*, those with $h_i < 0$ *upper*; as $P$ is bounded there are both. Let

$$f_i(y) = \frac{c_i - \langle y, \rho_i' \rangle}{h_i}, \qquad f = \max_{h_i > 0} f_i, \qquad g = \min_{h_i < 0} f_i,$$

let $\pi(x) = y$, and let $Q = \pi(P)$. Then $P = \{(y, s) : y \in Q,\ f(y) \leqslant s \leqslant g(y)\}$. For a lower facet $\Phi_i$ the map $\pi$ is injective on $\Phi_i$, so $D_i = \pi(\Phi_i)$ is a polygon whose vertices are the images of the vertices of $\Phi_i$, hence lie in $\mathbb{Z}^2$; moreover $D_i = \{y \in Q : f(y) = f_i(y)\}$, and the polygons $D_i$ cover $Q$, since $(y, f(y)) \in P$ for $y \in Q$. The same holds for the upper facets with $g$ in place of $f$; we write $U_j = \pi(\Phi_j)$ for them. Figure 3 shows the cells $D_i \cap U_j$ for a truncated cube.

**Proposition 7.** *Let $P$ be a lattice 3-polytope with normal fan $\Sigma$ and let $m \in \mathbb{Z}^3 \setminus \{0\}$ satisfy:*

*(P1) $\langle \rho_i, m \rangle \in \{-1, 0, 1\}$ for every ray $\rho_i$;*

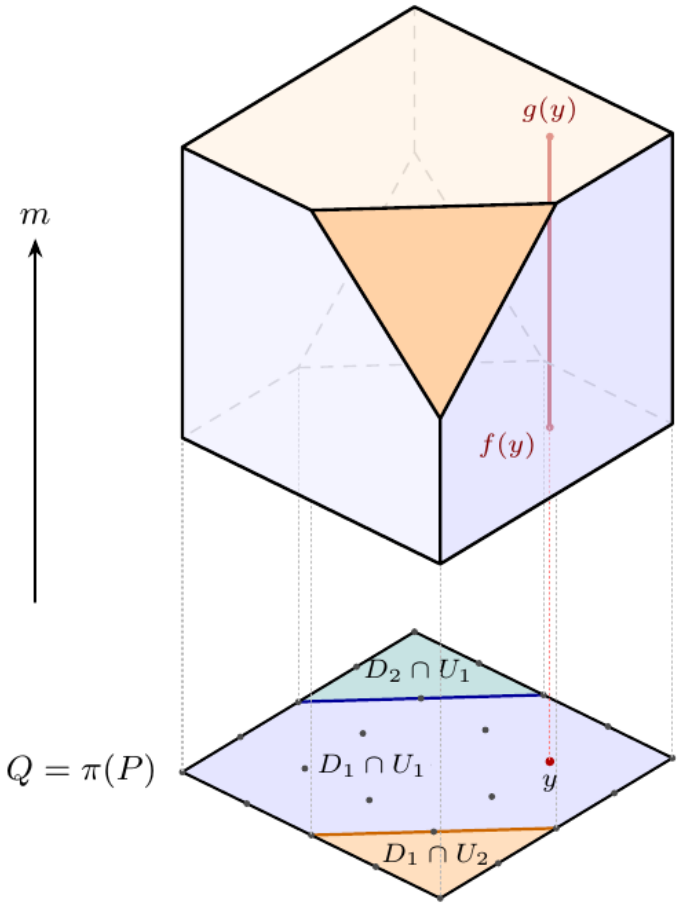



Figure 3: The projection along $m$. Over each point $y$ of the shadow $Q = \pi(P)$ the polytope is the segment from $f(y)$ to $g(y)$, and $Q$ is cut into the cells $D_i \cap U_j$ on which $f = f_i$ and $g = f_j$ are affine.

*(P2) for every two-dimensional cone* $\operatorname{cone}(\rho_a, \rho_b)$ *with* $h_a = h_b = 1$ *and every two-dimensional cone* $\operatorname{cone}(\rho_c, \rho_d)$ *with* $h_c = h_d = -1$,

$$(\rho_a - \rho_b) \times (\rho_c - \rho_d) \in \{0, m, -m\}.$$

*Then* $2P \cap \mathbb{Z}^3 = (P \cap \mathbb{Z}^3) + (P \cap \mathbb{Z}^3)$.

*Proof.* The rays span $\mathbb{R}^3$, so some height is nonzero, and by (P1) it is $\pm 1$; hence $m$ is primitive and the notation above applies. On $D_i$ we have $f = f_i$ and on $U_j$ we have $g = f_j$, and these functions take integer values on $\mathbb{Z}^2$ because $h_i, h_j = \pm 1$.

*Claim: the vertices of every polygon* $C = D_i \cap U_j$ *lie in* $\mathbb{Z}^2$. A vertex of $C$ that is a vertex of $D_i$ or of $U_j$ lies in $\mathbb{Z}^2$. Otherwise, by Lemma 4(b), it is the only common point of the lines spanned by an edge $e$ of $D_i$ and an edge $e'$ of $U_j$, and it lies in the relative interiors of both. The edge $e$ is the image of an edge $\Phi_i \cap \Phi_k$. If $h_k = 0$, then $e$ lies in the boundary line $\langle y, \rho_k' \rangle = c_k$ of $Q$. If $h_k = -1$, then $f \geqslant f_i = f_k \geqslant g \geqslant f$ on $e$, so

$f = g$ on $e$; since $g - f$ is concave and nonnegative on $Q$ and is not identically zero, it does not vanish at interior points of $Q$, so $e \subset \partial Q$. In both cases $e$ lies in an edge of $Q$, and $e' \subset Q$ cannot cross its line in a relative interior point of $e'$. Hence $h_k = 1$, and $e$ lies on the line $\langle y, \alpha' \rangle = c_i - c_k$ with $\alpha = \rho_i - \rho_k$. In the same way $e'$ is the image of $\Phi_j \cap \Phi_l$ with $h_j = h_l = -1$ and lies on the line $\langle y, \beta' \rangle = c_j - c_l$ with $\beta = \rho_j - \rho_l$. Let $b_1^*, b_2^*, m^*$ be the basis dual to $b_1, b_2, m$. As $\langle \alpha, m \rangle = \langle \beta, m \rangle = 0$, we have $\alpha = \alpha_1' b_1^* + \alpha_2' b_2^*$, similarly for $\beta$, and for every $w \in \mathbb{R}^3$

$$\langle \alpha \times \beta, w \rangle = \det(\alpha, \beta, w) = \det(\alpha', \beta') \det(b_1^*, b_2^*, m^*) \langle m, w \rangle.$$

Hence $\alpha \times \beta = \pm \det(\alpha', \beta')\, m$, and (P2) gives $|\det(\alpha', \beta')| \leqslant 1$. The two lines meet in one point, so $\det(\alpha', \beta') \neq 0$, and Lemma 4(a) shows that the vertex lies in $\mathbb{Z}^2$.

Now let $z = (Y, S) \in 2P \cap \mathbb{Z}^3$. Then $Y/2 \in Q$, so $Y/2 \in C = D_i \cap U_j$ for some $i, j$, and $2f(Y/2) \leqslant S \leqslant 2g(Y/2)$. By the claim and Lemma 3, $Y = y_1 + y_2$ with $y_1, y_2 \in C \cap \mathbb{Z}^2$. Since $f = f_i$ and $g = f_j$ are affine on $C$, $f(y_1) + f(y_2) = 2f(Y/2)$ and $g(y_1) + g(y_2) = 2g(Y/2)$. The four numbers $f(y_\nu), g(y_\nu)$ are integers with $f(y_\nu) \leqslant g(y_\nu)$, so there are integers $s_\nu \in [f(y_\nu), g(y_\nu)]$ with $s_1 + s_2 = S$. The points $(y_\nu, s_\nu)$ lie in $P \cap \mathbb{Z}^3$ and add up to $z$. □

**Proposition 8.** *Let $P$ be a lattice 3-polytope with normal fan $\Sigma$ and let $m \in \mathbb{Z}^3$. Suppose that exactly one ray $\rho_0$ has negative height, that $h_0 = -1$, and that $h_i \in \{0, 1, 2\}$ for all other rays. Then $2P \cap \mathbb{Z}^3 = (P \cap \mathbb{Z}^3) + (P \cap \mathbb{Z}^3)$.*

*Proof.* As $h_0 = -1$, $m$ is primitive. The only upper facet is $\Phi_0$, so $g = f_0$ is affine on all of $\mathbb{R}^2$ and takes integer values on $\mathbb{Z}^2$. Let $z = (Y, S) \in 2P \cap \mathbb{Z}^3$. Then $Y/2 \in D_i$ for a lower facet $\Phi_i$, and $2f(Y/2) \leqslant S \leqslant 2g(Y/2)$. We claim that there are $y_1, y_2 \in D_i \cap \mathbb{Z}^2$ with $y_1 + y_2 = Y$ and $\lceil f(y_1) \rceil + \lceil f(y_2) \rceil \leqslant S$. Given the claim, $\lceil f(y_\nu) \rceil \leqslant g(y_\nu)$ because $g(y_\nu)$ is an integer, and $S \leqslant 2g(Y/2) = g(y_1) + g(y_2)$; so there are integers $s_\nu \in [\lceil f(y_\nu) \rceil, g(y_\nu)]$ with $s_1 + s_2 = S$, and $(y_1, s_1) + (y_2, s_2) = z$ with both summands in $P \cap \mathbb{Z}^3$.

If $h_i = 1$, then $f = f_i$ takes integer values on $\mathbb{Z}^2$, and Lemma 3 gives $y_1 + y_2 = Y$ in $D_i \cap \mathbb{Z}^2$ with $f(y_1) + f(y_2) = 2f(Y/2) \leqslant S$. Let $h_i = 2$. As $\rho_i$ is primitive, $\rho_i' \notin 2\mathbb{Z}^2$, so $\Lambda = \{y \in \mathbb{Z}^2 : \langle y, \rho_i' \rangle \text{ even}\}$ has index 2 in $\mathbb{Z}^2$, and $T = \{y \in \mathbb{Z}^2 : \langle y, \rho_i' \rangle \equiv c_i \pmod 2\}$ is a coset of $\Lambda$. For $y \in \mathbb{Z}^2$ the number $f_i(y)$ is an integer if $y \in T$ and lies in $\mathbb{Z} + \frac{1}{2}$ otherwise. A vertex $(y, s)$ of $\Phi_i$ satisfies $\langle y, \rho_i' \rangle + 2s = c_i$, so the vertices of $D_i$ lie in $T$.

If $\langle Y, \rho_i' \rangle$ is even, choose $w \in T$; then $Y - 2w \in \Lambda$, and Lemma 3 for the lattice $\Lambda$ and the polygon $D_i - w$ gives $Y - 2w = y_1' + y_2'$ with $y_\nu' \in (D_i - w) \cap \Lambda$. The points $y_\nu = y_\nu' + w$ lie in $D_i \cap T$, so $f(y_\nu) \in \mathbb{Z}$ and $f(y_1) + f(y_2) = 2f(Y/2) \leqslant S$.

If $\langle Y, \rho_i' \rangle$ is odd, Lemma 3 gives $Y = y_1 + y_2$ with $y_\nu \in D_i \cap \mathbb{Z}^2$. As $\langle y_1, \rho_i' \rangle + \langle y_2, \rho_i' \rangle$ is odd, exactly one of $y_1, y_2$ lies in $T$, so $\lceil f(y_1) \rceil + \lceil f(y_2) \rceil = f(y_1) + f(y_2) + \frac{1}{2} = 2f(Y/2) + \frac{1}{2}$. Moreover $2f(Y/2) = c_i - \frac{1}{2}\langle Y, \rho_i' \rangle \in \mathbb{Z} + \frac{1}{2}$, and $S \geqslant 2f(Y/2)$ is an integer, so $S \geqslant 2f(Y/2) + \frac{1}{2}$. □

Propositions 7 and 8 do not use smoothness. We write $S(u)$, $P(m)$ and $Q(m)$ for a vector satisfying the hypotheses of Proposition 6, 7 or 8, respectively, and call it a *certificate* of $\Sigma$.

## 4 The fans

A smooth complete fan in $\mathbb{R}^3$ is the normal fan of a smooth 3-polytope exactly when its toric variety is projective, and the number of facets of the polytope is the number of rays of the fan. If $\Sigma$ is a smooth fan and $\tau = \mathrm{cone}(\rho_{i_1}, \ldots, \rho_{i_d})$ is a cone of $\Sigma$ of dimension $d \in \{2, 3\}$, the *star subdivision* of $\Sigma$ at $\tau$ has the additional ray $\rho_{i_1} + \cdots + \rho_{i_d}$; each maximal cone $\sigma \supseteq \tau$ is replaced by the cones spanned by the new ray and the faces of $\sigma$ of dimension 2 that do not contain $\tau$. It is the fan of the blowup of the toric variety along the torus-invariant point ($d = 3$) or curve ($d = 2$) given by $\tau$ [4, Section 3.3]. We write $p_{ijk}$ for the star subdivision at $\mathrm{cone}(\rho_i, \rho_j, \rho_k)$ and $c_{ij}$ for the one at $\mathrm{cone}(\rho_i, \rho_j)$, and we give the new ray the next free index.

The following fans have rays depending on integer parameters $a, b \in \mathbb{Z}$ and $p \in \mathbb{Z}^4$ or $p \in \mathbb{Z}^5$; cones are listed by the indices of their rays.

- $Z_a$: $\rho_0 = (1, 0, 0)$, $\rho_1 = (0, 1, 0)$, $\rho_2 = (-1, -1, a)$, $\rho_3 = (0, 0, 1)$, $\rho_4 = (0, 0, -1)$; maximal cones 013, 014, 123, 124, 023, 024. This is the fan of $\mathbb{P}(\mathcal{O} \oplus \mathcal{O}(a))$ over $\mathbb{P}^2$.

- $W_{a,b}$: $\rho_0 = (1, 0, 0)$, $\rho_1 = (0, 1, 0)$, $\rho_2 = (-1, -1, 0)$, $\rho_3 = (0, 0, 1)$, $\rho_4 = (a, b, -1)$, with the same maximal cones. This is the fan of $\mathbb{P}(\mathcal{O} \oplus \mathcal{O}(a) \oplus \mathcal{O}(b))$ over $\mathbb{P}^1$.

- $F_{a,p}$: $\rho_i = (v_i, p_i)$ for $0 \leqslant i \leqslant 3$, where $v_0 = (1, 0)$, $v_1 = (0, 1)$, $v_2 = (-1, a)$, $v_3 = (0, -1)$ are the rays of the Hirzebruch surface, $\rho_4 = (0, 0, 1)$, $\rho_5 = (0, 0, -1)$; maximal cones $\{i, i + 1, 4\}$ and $\{i, i + 1, 5\}$ with $i + 1$ taken modulo 4. This is the fan of a $\mathbb{P}^1$-bundle over the Hirzebruch surface.

- $G^j_{a,p}$ for $0 \leqslant j \leqslant 3$: the fan built in the same way over the five vectors $v_0, \ldots, v_j, v_j + v_{j+1}, v_{j+1}, \ldots, v_3$ (indices modulo 4), numbered $0, \ldots, 4$ in this order, with $\rho_5 = (0, 0, 1)$ and $\rho_6 = (0, 0, -1)$. These are $\mathbb{P}^1$-bundles over the blowups of the Hirzebruch surface at a fixed point.

- $C_a$: $\rho_0 = (1, 0, 0)$, $\rho_1 = (0, 1, 0)$, $\rho_2 = (0, 0, 1)$, $\rho_3 = (0, -1, a)$, $\rho_4 = (0, 0, -1)$, $\rho_5 = (0, 1, -1)$, $\rho_6 = (-1, 2, -1)$; maximal cones 012, 023, 034, 045, 056, 016, 126, 236, 346, 456. This is the family (a) of [10, Section 3].

**Proposition 9.** *Let $\Sigma$ be a smooth complete fan in $\mathbb{R}^3$ with seven rays whose toric variety is projective. Then $\Sigma$ is equivalent under $\mathrm{GL}_3(\mathbb{Z})$ to one of the following fans:*

*(i) a star subdivision, at a cone of dimension 2 or 3, of a star subdivision of $Z_a$ or of $W_{a,b}$ at a cone of dimension 2 or 3 (600 families);*

*(ii) a star subdivision of $F_{a,p}$ at a cone of dimension 2 or 3 (20 families);*

*(iii) $G^j_{a,p}$ with $0 \leqslant j \leqslant 3$ (4 families);*

*(iv) $C_a$ (1 family).*

*Proof.* Let $X$ be the toric variety of $\Sigma$. It is smooth and projective of Picard number $7-3=4$ [4, Chapter 4]. Call a smooth complete toric threefold *minimal* if its fan is not a star subdivision of a smooth complete fan at a cone of dimension 2 or 3, that is, if it is not the blowup of a smooth complete toric threefold along a torus-invariant point or curve. By the classification of Miyake and Oda [15, Section 9] and Nagaya [14], in the form stated in [10, Section 3, proof of the Lemma], every minimal smooth complete toric threefold of Picard number less than 5 is $\mathbb{P}^3$, a $\mathbb{P}^2$-bundle over $\mathbb{P}^1$, a $\mathbb{P}^1$-bundle over a smooth projective toric surface, the variety of the fan $C_a$, or one further variety of Picard number 4, which is not projective [15, Proposition 9.4].

If $X$ is minimal, it is not $\mathbb{P}^3$ or a $\mathbb{P}^2$-bundle over $\mathbb{P}^1$, whose Picard numbers are 1 and 2. If it is a $\mathbb{P}^1$-bundle over a surface $S$, then $S$ has Picard number 3, so its fan has five rays and $S$ is the blowup of a Hirzebruch surface at a fixed point [4, Theorem 10.4.3]; the fan of $X$ is then $G^j_{a,p}$ for some $j, a, p$. Otherwise $\Sigma$ is $C_a$.

If $X$ is not minimal, $\Sigma$ is a star subdivision of a smooth complete fan $\Sigma'$ with six rays. If the variety of $\Sigma'$ is minimal, it has Picard number 3 and by the same classification it is a $\mathbb{P}^1$-bundle over a surface with four rays, that is, over a Hirzebruch surface; so $\Sigma'$ is $F_{a,p}$. Otherwise $\Sigma'$ is a star subdivision of a smooth complete fan with five rays, which is $Z_a$ or $W_{a,b}$ by Kleinschmidt's classification of smooth complete fans with $n+2$ rays [12].

The fans $Z_a$ and $W_{a,b}$ have 6 maximal cones and 9 two-dimensional cones, and each of their star subdivisions has 8 maximal cones and 12 two-dimensional cones; this gives $2 \cdot 15 \cdot 20 = 600$ families in (i) and $8 + 12 = 20$ in (ii). □

Figure 4 shows the two kinds of star subdivision and how the four kinds of families arise.

## 5 Certificates

**Proposition 10.** *For each of the* 625 *families of Proposition 9, the vector listed in Appendix A is a certificate of every fan of the family, for all integer values of the parameters.*

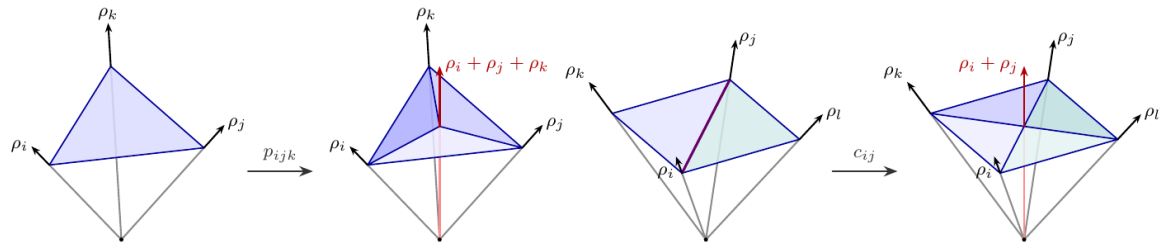
We first treat the 25 families (ii)–(iv) by hand, then give two examples from (i), and then describe the verification of all 625 families by computer.

**Lemma 11.** *Let $v_0, \dots, v_{q-1}$ be the rays, in cyclic order, of a smooth complete fan in $\mathbb{R}^2$, let $p \in \mathbb{Z}^q$, and let $\Sigma$ be the fan with rays $\rho_i = (v_i, p_i)$ for $0 \leqslant i < q$, $\rho_q = (0,0,1)$, $\rho_{q+1} = (0,0,-1)$, and maximal cones* $\operatorname{cone}(\rho_i, \rho_{i+1}, \rho_q)$ *and* $\operatorname{cone}(\rho_i, \rho_{i+1}, \rho_{q+1})$*, with $i+1$ taken modulo $q$. Let $u = (0,0,-1)$.*

*(a) If $q \leqslant 5$, then $u$ satisfies (S1) and (S2) for $\Sigma$.*

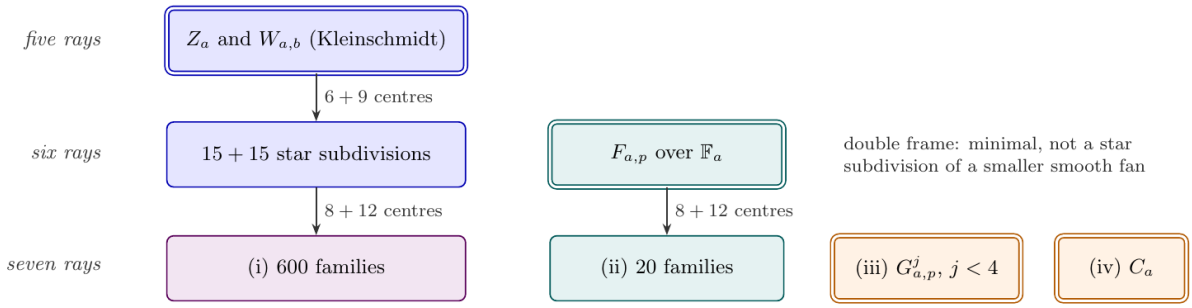
*(b) If $q \leqslant 4$, then $u$ satisfies (S1) and (S2) for every star subdivision of $\Sigma$ at a cone of dimension 2 or 3.*

*Proof.* For $x, y \in \mathbb{R}^3$ with first two coordinates $\bar{x}, \bar{y}$ we have $\det(u, x, y) = -\det(\bar{x}, \bar{y})$. Hence $\rho_q$ and $\rho_{q+1}$ are isolated in $G_u$, and $i < q$ and $j < q$ are adjacent exactly when




(a) blowup of a point: one cone becomes three

(b) blowup of a curve: two cones become four




(c) the families of Proposition 9 by number of rays

Figure 4: (a) The star subdivision $p_{ijk}$ of a maximal cone, the blowup of a torus-fixed point. (b) The star subdivision $c_{ij}$ of a two-dimensional cone, the blowup of a torus-invariant curve. (c) The families of Proposition 9: the fans with five rays are $Z_a$ and $W_{a,b}$, each has 15 star subdivisions with six rays, and each of these has 20 further star subdivisions; together with the subdivisions of $F_{a,p}$ and the minimal fans $G^j_{a,p}$ and $C_a$ this gives $600 + 20 + 4 + 1 = 625$ families.

$|\det(v_i, v_j)| \geqslant 2$. Consecutive rays $v_i, v_{i+1}$ have $|\det(v_i, v_{i+1})| = 1$, so the restriction of $G_u$ to $\{0, \dots, q-1\}$ is contained in the complement of the cycle $0, 1, \dots, q-1$. For $q = 3, 4, 5$ this complement is empty, a perfect matching, and a cycle of length five, and none of them contains a cycle of length four.

(a) Every maximal cone contains $\rho_q$ or $\rho_{q+1}$, and $u = -\rho_q = \rho_{q+1}$, so (S1) holds; (S2) was just shown.

(b) Let $w$ be the new ray. If the subdivided cone is $\operatorname{cone}(\rho_i, \rho_{i+1}, \rho_q)$, then $w = \rho_i + \rho_{i+1} + \rho_q$ and the new maximal cones are $\operatorname{cone}(\rho_i, \rho_{i+1}, w)$, in which $u = \rho_i + \rho_{i+1} - w$, and two cones containing $\rho_q$. If it is $\operatorname{cone}(\rho_i, \rho_{i+1})$, all new maximal cones contain $\rho_q$ or $\rho_{q+1}$. If it is $\operatorname{cone}(\rho_i, \rho_q)$, then $w = \rho_i + \rho_q$ and the new maximal cones without $\rho_q$ are $\operatorname{cone}(\rho_{i-1}, \rho_i, w)$ and $\operatorname{cone}(\rho_i, \rho_{i+1}, w)$, in which $u = \rho_i - w$. The cones containing $\rho_{q+1}$ are symmetric. So (S1) holds in all cases. For (S2), in the first two cases $\bar{w} = v_i + v_{i+1}$, and $v_0, \dots, v_i, v_i + v_{i+1}, v_{i+1}, \dots, v_{q-1}$ are the rays, in cyclic order, of a smooth complete fan in $\mathbb{R}^2$ with $q + 1 \leqslant 5$ rays; the argument above applies. In the third case $\bar{w} = v_i$, so $w$ is adjacent in $G_u$ exactly to the neighbours of $i$ and not to $i$. A cycle of length four would have to pass through $w$, since the graph without $w$ has none, and then $i$ would have two

distinct neighbours; but for $q \leqslant 4$ every vertex has at most one neighbour. □

Lemma 11(b) with $q = 4$ gives the certificate $S(0, 0, -1)$ for the 20 families (ii), and Lemma 11(a) with $q = 5$ gives it for the four families $G^j_{a,p}$. The fans $G^j_{a,p}$ are pseudo-prisms in the sense of [5], whose Theorem 4.6 also covers them. For $C_a$ take $m = (-1, -1, 0)$: the heights of $\rho_0, \dots, \rho_6$ are $-1, -1, 0, 1, 0, -1, -1$, so (P1) holds, and only $\rho_3$ has height 1, so no two-dimensional cone has both heights 1 and (P2) holds trivially. This is the certificate $P(-1, -1, 0)$.

**Two examples from (i).** Subdividing $Z_a$ at $p_{013}$ and then at $p_{015}$ adds $\rho_5 = (1, 1, 1)$ and $\rho_6 = \rho_0 + \rho_1 + \rho_5 = (2, 2, 1)$. For $m = (-1, 1, 0)$ the heights of $\rho_0, \dots, \rho_6$ are $-1, 1, 0, 0, 0, 0, 0$, so (P1) holds and (P2) is vacuous: this is $P(-1, 1, 0)$. Subdividing $W_{a,b}$ at $c_{24}$ and then at $c_{45}$ adds $\rho_5 = \rho_2 + \rho_4 = (a - 1, b - 1, -1)$ and $\rho_6 = \rho_4 + \rho_5 = (2a - 1, 2b - 1, -2)$. For $m = (0, 0, -1)$ the heights are $0, 0, 0, -1, 1, 1, 2$, which is the certificate $Q(0, 0, -1)$ of Proposition 8.

**Verification of all families.** In each family the rays are vectors whose coordinates are integer affine functions of the parameters. Every quantity in the hypotheses of Propositions 6–8 is then a polynomial in the parameters: the coordinates of $u$ in the basis of a maximal cone (the cone determinants are the constants $\pm 1$), the determinants $\det(u, \rho_i, \rho_j)$, the heights $\langle \rho_i, m \rangle$ and the cross products in (P2). A certificate is accepted only if each quantity that must lie in a finite set is a constant polynomial whose value lies in that set. For (S2), a pair $i, j$ is counted as adjacent unless $\det(u, \rho_i, \rho_j)$ is a constant of absolute value at most 1; the graph obtained contains $G_u$ for every value of the parameters, and it is required to contain no cycle of length four. An accepted certificate therefore satisfies the hypotheses for all values of the parameters.

The check is formalized in Lean 4 [8] with Mathlib [13]. The formalization builds the 625 families of Proposition 9 from the five base fans by the star subdivisions of Section 4, states the hypotheses of Propositions 6, 7 and 8 as propositions about integer vectors, and proves that every fan of every family, at every value of the parameters, has a certificate (Appendix B). Decision procedures compute on the polynomial coefficients of the rays as described above, and soundness theorems show that a procedure that accepts gives the hypothesis at every value of the parameters. The procedures are evaluated by the kernel itself, without compiled code; the proof has no unproved step and depends only on the axioms of Lean's logic, namely propositional extensionality, choice and the soundness of quotients.

Four independent programs repeat the check on the same table. Two of them compute with exact polynomials in the parameters. The other two use that every quantity above has total degree at most 3, hence degree at most 3 in each parameter, so it equals a constant $c$ if and only if it takes the value $c$ at every point of a grid with four values per parameter; they use the grids $\{0, 1, 2, 3\}$ and $\{-2, -1, 1, 3\}$. Each of the five checks also verifies that every fan of the families is smooth and complete and that every family has exactly one certificate in the table.

## 6 Proof of Theorem 1 and Corollary 2

*Proof of Theorem 1.* Let $P \subset \mathbb{R}^3$ be a smooth lattice polytope with seven facets. Its normal fan $\Sigma$ is a smooth complete fan with seven rays whose toric variety is projective. By Proposition 9 there is $\gamma \in \mathrm{GL}_3(\mathbb{Z})$ such that $\gamma\Sigma$ is a fan of one of the 625 families, for some values of the parameters. The polytope $P^\gamma = (\gamma^{-1})^{\mathsf{T}} P$ is a smooth lattice polytope with normal fan $\gamma\Sigma$, and the map $x \mapsto (\gamma^{-1})^{\mathsf{T}} x$ preserves $\mathbb{Z}^3$, so $P$ has the IDP if and only if $P^\gamma$ has. By Proposition 10, $\gamma\Sigma$ has a certificate, and Proposition 6, 7 or 8 gives $2P^\gamma \cap \mathbb{Z}^3 = (P^\gamma \cap \mathbb{Z}^3) + (P^\gamma \cap \mathbb{Z}^3)$. By Lemma 3, $P^\gamma$ has the IDP. □

*Proof of Corollary 2.* The normal fan of a smooth lattice 3-polytope with at most six facets is a smooth complete fan in $\mathbb{R}^3$ with at most $3 + 3$ rays, so the polytope has the IDP by [19, Theorem 1.4]. Seven facets is Theorem 1. □

The criteria of Section 3 do not settle every smooth 3-polytope: they are sufficient conditions on the normal fan, and the case of eight or more facets remains open.

## A The certificates

For the families (i) we list the two subdivisions in the order in which they are made; the new rays are $\rho_5$ and $\rho_6$. For each family the certificate is $S(u)$, $P(m)$ or $Q(m)$ in the coordinates of Section 4. The list below is printed from the same table that the formal proof and the four programs check.

**Double blowups of $Z_a$ (300 pairs).** The certificate is $S(0, 0, -1)$ for 264 pairs; the other pairs are:

- $P(-1, 0, 0)$ (20 pairs): $p_{013}\,c_{15}$, $p_{014}\,c_{15}$, $p_{023}\,p_{025}$, $p_{023}\,c_{05}$, $p_{023}\,c_{25}$, $p_{024}\,p_{025}$, $p_{024}\,c_{05}$, $p_{024}\,c_{25}$, $p_{123}\,c_{15}$, $p_{124}\,c_{15}$, $c_{03}\,p_{025}$, $c_{04}\,p_{025}$, $c_{13}\,p_{015}$, $c_{13}\,p_{125}$, $c_{13}\,c_{15}$, $c_{14}\,p_{015}$, $c_{14}\,p_{125}$, $c_{14}\,c_{15}$, $c_{23}\,p_{025}$, $c_{24}\,p_{025}$.
- $P(-1, 1, 0)$ (12 pairs): $p_{013}\,p_{015}$, $p_{013}\,c_{05}$, $p_{014}\,p_{015}$, $p_{014}\,c_{05}$, $p_{123}\,c_{25}$, $p_{124}\,c_{25}$, $c_{03}\,p_{015}$, $c_{04}\,p_{015}$, $c_{23}\,p_{125}$, $c_{23}\,c_{25}$, $c_{24}\,p_{125}$, $c_{24}\,c_{25}$.
- $P(0, -1, 0)$ (4 pairs): $p_{123}\,p_{125}$, $p_{124}\,p_{125}$, $c_{03}\,c_{05}$, $c_{04}\,c_{05}$.

**Double blowups of $W_{a,b}$ (300 pairs).** The certificate is $S(-1, -1, 0)$ for 186 pairs; the other pairs are:

- $P(0, 0, -1)$ (72 pairs): $p_{013}\,p_{014}$, $p_{013}\,p_{024}$, $p_{013}\,p_{124}$, $p_{013}\,c_{04}$, $p_{013}\,c_{14}$, $p_{013}\,c_{24}$, $p_{014}\,p_{013}$, $p_{014}\,p_{023}$, $p_{014}\,p_{123}$, $p_{014}\,c_{03}$, $p_{014}\,c_{13}$, $p_{014}\,c_{23}$, $p_{023}\,p_{014}$, $p_{023}\,p_{024}$, $p_{023}\,p_{124}$, $p_{023}\,c_{04}$, $p_{023}\,c_{14}$, $p_{023}\,c_{24}$, $p_{024}\,p_{013}$, $p_{024}\,p_{023}$, $p_{024}\,p_{123}$, $p_{024}\,c_{03}$, $p_{024}\,c_{13}$, $p_{024}\,c_{23}$, $p_{123}\,p_{014}$, $p_{123}\,p_{024}$, $p_{123}\,p_{124}$, $p_{123}\,c_{04}$, $p_{123}\,c_{14}$, $p_{123}\,c_{24}$, $p_{124}\,p_{013}$, $p_{124}\,p_{023}$, $p_{124}\,p_{123}$, $p_{124}\,c_{03}$, $p_{124}\,c_{13}$, $p_{124}\,c_{23}$, $c_{03}\,p_{014}$, $c_{03}\,p_{024}$, $c_{03}\,p_{124}$, $c_{03}\,c_{04}$, $c_{03}\,c_{14}$, $c_{03}\,c_{24}$, $c_{04}\,p_{013}$, $c_{04}\,p_{023}$, $c_{04}\,p_{123}$, $c_{04}\,c_{03}$, $c_{04}\,c_{13}$, $c_{04}\,c_{23}$, $c_{13}\,p_{014}$, $c_{13}\,p_{024}$, $c_{13}\,p_{124}$, $c_{13}\,c_{04}$, $c_{13}\,c_{14}$, $c_{13}\,c_{24}$, $c_{14}\,p_{013}$, $c_{14}\,p_{023}$, $c_{14}\,p_{123}$, $c_{14}\,c_{03}$, $c_{14}\,c_{13}$, $c_{14}\,c_{23}$, $c_{23}\,p_{014}$, $c_{23}\,p_{024}$, $c_{23}\,p_{124}$, $c_{23}\,c_{04}$, $c_{23}\,c_{14}$, $c_{23}\,c_{24}$, $c_{24}\,p_{013}$, $c_{24}\,p_{023}$, $c_{24}\,p_{123}$, $c_{24}\,c_{03}$, $c_{24}\,c_{13}$, $c_{24}\,c_{23}$.

- $S(-1,0,0)$ (22 pairs): $p_{013}\,p_{035}$, $p_{014}\,p_{045}$, $p_{023}\,p_{035}$, $p_{023}\,c_{05}$, $p_{024}\,p_{045}$, $p_{024}\,c_{05}$, $p_{123}\,c_{15}$, $p_{124}\,c_{15}$, $c_{02}\,p_{035}$, $c_{02}\,p_{045}$, $c_{02}\,c_{05}$, $c_{03}\,c_{13}$, $c_{04}\,c_{14}$, $c_{12}\,p_{135}$, $c_{12}\,p_{145}$, $c_{12}\,c_{15}$, $c_{13}\,p_{035}$, $c_{13}\,c_{03}$, $c_{14}\,p_{045}$, $c_{14}\,c_{04}$, $c_{23}\,p_{035}$, $c_{24}\,p_{045}$.
- $S(0,-1,0)$ (8 pairs): $p_{013}\,p_{135}$, $p_{014}\,p_{145}$, $p_{123}\,p_{135}$, $p_{124}\,p_{145}$, $c_{03}\,p_{135}$, $c_{04}\,p_{145}$, $c_{23}\,p_{135}$, $c_{24}\,p_{145}$.
- $Q(0,0,-1)$ (6 pairs): $p_{014}\,c_{45}$, $p_{024}\,c_{45}$, $p_{124}\,c_{45}$, $c_{04}\,c_{45}$, $c_{14}\,c_{45}$, $c_{24}\,c_{45}$.
- $Q(0,0,1)$ (6 pairs): $p_{013}\,c_{35}$, $p_{023}\,c_{35}$, $p_{123}\,c_{35}$, $c_{03}\,c_{35}$, $c_{13}\,c_{35}$, $c_{23}\,c_{35}$.

**Blowups of $F_{a,p}$ (20 centres).** The certificate is $S(0,0,-1)$ for every centre.

**The fans $G^j_{a,p}$.** $G^0$: $S(0,0,-1)$; $G^1$: $S(0,0,-1)$; $G^2$: $S(0,0,-1)$; $G^3$: $S(0,0,-1)$.

**The fan $C_a$.** $P(-1,-1,0)$.

# B The formal statement

In the formalization a fan with integer rays is a list of rays together with a list of maximal cones, each given by three ray indices, and a family is the same data with rays whose coordinates are integer affine functions of a parameter vector $p$. For a family $X$ and a parameter vector $p$ let $X(p)$ be its fan at $p$; for a fan or family $B$ let $\mathcal{C}(B)$ be its list of cones of dimension 2 and 3, and let $B_c$ be the star subdivision of $B$ at $c \in \mathcal{C}(B)$, with the new ray as the last ray. A fan $F$ *has a certificate* if some $v \in \mathbb{Z}^3$ satisfies the hypotheses of Proposition 6, 7 or 8 for $F$; condition (S2) is stated as: for all ray indices $i,j,k,l$ with $i \neq k$ and $j \neq l$, the four pairs $(i,j)$, $(j,k)$, $(k,l)$, $(l,i)$ are not all adjacent in $G_u$. The theorem proved in Lean is the conjunction of the following five statements.

1. For every $c_1 \in \mathcal{C}(Z)$, every $c_2 \in \mathcal{C}(Z_{c_1})$ and every $p$, the fan $(Z_{c_1})_{c_2}(p)$ has a certificate.
2. For every $c_1 \in \mathcal{C}(W)$, every $c_2 \in \mathcal{C}(W_{c_1})$ and every $p$, the fan $(W_{c_1})_{c_2}(p)$ has a certificate.
3. For every $c \in \mathcal{C}(F)$ and every $p$, the fan $F_c(p)$ has a certificate.
4. For every $j < 4$ and every $p$, the fan $G^j(p)$ has a certificate.
5. For every $p$, the fan $C(p)$ has a certificate.

Here $Z$, $W$, $F$, $G^j$ and $C$ are the families $Z_a$, $W_{a,b}$, $F_{a,p}$, $G^j_{a,p}$ and $C_a$ of Section 4. The data of the formal proof are generated from the table of Appendix A, and the build of the proof compares them with that table.

## Data availability

Deep Bhattacharjee computed the Lean formalization and the Julia, C, C++, shell and Python computations with the help of AI assistance (Claude, Anthropic). The Lean formalization of the proof in this paper is archived at Zenodo [2], doi:10.5281/zenodo.23329590. The paper is available at https://github.com/creelie/probable-octo-palm-tree/blob/main/preprintOdaConjectureSevenFacets/OdaConjectureSevenFacets.pdf.


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